

KARINA GREWAL

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EDUCATION

University of Illinois Urbana-Champaign [MS in Statistics]

Aug 2024 - Dec 2025

University of British Columbia [BS in Statistics]

Sept 2018 - May 2023

PROFESSIONAL EXPERIENCE

Data Analytics Consultant Intern

Chicago, IL

Health Care Service Corporation (HCSC)

June 2025 - Aug 2025

- Assisted with data sourcing, cleaning, and integration to manage real-world large scale healthcare data using intermediate SQL and Python; offered a new method to decrease running time for SQL queries by 90%
- Conducted causal inference estimated Average Treatment Effects (ATE) ensuring data are robust, and interpretable
- Performed variable selection and feature reduction using Variance Inflation Factor (VIF) and Recursive Feature Elimination (RFE) to improve model interpretability and performance while working independently
- Delivered key findings and model-driven insights to senior leadership, driving impactful results

Statistics Research Consultant

Vancouver, BC

Case Studies in Statistics

Jan 2023 - April 2023

- Partnered with a UBC physician as to analyze outcomes from 26 surgical mice, contributing to a peer-reviewed publication [*Canadian Journal of Surgery* (Dec 2023)]
- Applied survival analysis, kaplan-meier analysis, logistic regression, and nonparametric tests in R, showing that perioperative antibiotics reduced bacterial colony counts by ~99% and improved healing odds by 87.5%
- Produced 20+ publication-ready ggplot2 visualizations and presented findings to clinical stakeholders, driving statistical insights into impactful results

SELECTED EXPERIENCE

Power Consumption in Morocco

Champaign, IL

Time Series Analysis

Jan 2025 - May 2025

- Analyzed ~350k electricity consumption records from Tetouan, Morocco (2017) to forecast regional energy demand using weather and temporal features; Designed metrics frameworks to measure energy efficiency
- Preprocessed and engineered time-series features; experimented with linear regression models, transitioned to SARIMA for improved forecasting, and benchmarked against XGBoost with time-series CV
- Conducted spectral analysis to uncover hidden seasonal and cyclical patterns

Classifying Brain Tumors

Champaign, IL

Basics of Statistical Machine Learning

Jan 2025 - May 2025

- Collaborated on a project to classify glioma tumors using TCGA clinical and genetic data
- Conducted exploratory data analysis on 862 patients; contributed to implementing and tuning classification models (KNN, Logistic Regression, Random Forest, SVM) to generate data-driven recommendations
- Delivered models achieving up to 0.92 AUC / 87% accuracy, identifying IDH1 mutation and age at diagnosis as key predictors to support clinical research

LEADERSHIP EXPERIENCE

Statistics Consulting Club (STATcom) Executive [Champaign, IL]

May 2025 - Dec 2025

- Led a consulting project for a community client aimed at expanding a mentorship program
- Designed surveys, managed data collection, and built dashboards to assess mentorship program outcomes
- Analyzed survey data and built clear visualizations to demonstrate program impact, delivering insights and recommendations to non-technical stakeholders to guide program expansion

SKILLS

- Proficient with: Python (pandas, scikit-learn, CausalML, FLAML), SQL, R, Databricks, Tableau, Power BI, Excel (advanced), LaTeX, GitHub, communicating insights to technical and non-technical audiences
- Analytics/Techniques: GLMs, CART, time-series forecasting, causal inference, classification, residual analysis, A/B testing, KPI tracking, ETL/data pipelines, regex query writing